

NS-3 Project Report

Md. Afham Adian

ID: 2105119

April 7, 2026

a. Network topologies under simulation

Base: TCPWestwoodPlus

Paper's Modification: TCPWestwoodPlusNew

My Modification: TCPDualModifiedTCPWestwoodPlusNew

Wired topology

The wired setup uses a n -node multi-flow dumbbell topology with one bottleneck link:

$$N0 \leftrightarrow N1 \leftrightarrow N2 \leftrightarrow N3,$$

with two $N0$ being sender and $N3$ being receivers. Multiple TCP flows share the same bottleneck path segment ($N1$ – $N2$), enabling congestion-control comparison under contention.

For OAT sweeps with larger node counts, a canonical dumbbell is used:

- Total nodes: $n_nodes = 2 \times n_leaf + 2$
- Left side leaves: senders
- Right side leaves: receivers
- Middle 2 nodes: routers connected by the single bottleneck link

Example: for $n_nodes = 60$, $n_leaf = 29$ on each side and 2 core router nodes.

Wireless topology (IEEE 802.15.4 static)

The wireless setup uses a static IEEE 802.15.4 network with 6LoWPAN adaptation and IPv6 stack. Nodes are placed on a square grid using a constant-position mobility model (no movement during simulation).

Topology characteristics:

- PHY/MAC: IEEE 802.15.4 (LR-WPAN)
- Adaptation: 6LoWPAN over LR-WPAN devices
- Network layer: IPv6
- Placement: static grid over a square area
- Area scaling: one side varied as $\{1, 2, 3, 4, 5\} \times Tx_range$

Traffic is configured as multiple TCP flows between selected source–destination node pairs within the static grid.

Meaning of static wireless:

- Node positions are fixed for the entire simulation (no mobility update over time).
- The connectivity changes only due to contention, load, protocol behavior, and packet errors, not due to node movement.
- Coverage-area variation is emulated by changing the square deployment area size ($1 \times$ to $5 \times Tx_range$), while keeping nodes static inside that area.

b. Parameters under variation

The simulations were conducted for two network categories:

- Wired network
- Wireless IEEE 802.15.4 (static)

The following parameters were varied according to the project checklist:

1. Number of nodes: $\{20, 40, 60, 80, 100\}$
2. Number of flows: $\{10, 20, 30, 40, 50\}$
3. Packet generation rate (packets per second): $\{100, 200, 300, 400, 500\}$
4. Coverage area multiplier (static wireless only): $\{1, 2, 3, 4, 5\} \times Tx_range$

For runtime optimization and controlled analysis, an **one-factor-at-a-time (OAT)** approach was used around the baseline point:

$$(n_nodes, n_flows, pps, coverage) = (60, 30, 300, 2 \times Tx_range).$$

Thus, each dimension was varied independently while keeping the others fixed at baseline values. TCP variants compared throughout the experiments were:

- `TcpWestwoodPlus`
- `TcpWestwoodPlusNew`
- `TcpDualModifiedTcpWestwoodPlusNew`

c. Modifications made in the simulator

The simulator was modified and extended in the following way.

Wired simulation updates

- A dedicated wired OAT simulation pipeline was added for automated batch runs and reproducible outputs.
- A dedicated 40-second wired time-series simulator was added to compare the three TCP variants over time.
- Time-series metrics were instrumented and exported as data files:
 - Throughput vs time
 - Delay vs time (cumulative delay sum)
 - Packet loss vs time
 - Congestion window (CWND) vs time
 - Bottleneck queue size vs time
 - RTT vs time
- Focused RTT/queue comparison plots were also generated for selected algorithm pairs.

Wireless simulation updates (IEEE 802.15.4 static)

- A static wireless simulation setup was aligned with IEEE 802.15.4 requirements.
- Wireless OAT automation scripts were added to run parameter sweeps consistently with the wired workflow.
- The required wireless metric set was exported per run:
 - Throughput
 - End-to-end delay
 - Packet delivery ratio (PDR)
 - Packet drop ratio

- Energy consumption for wireless nodes
- Coverage-area sweep support was included for static wireless scenarios using multipliers of Tx_range.

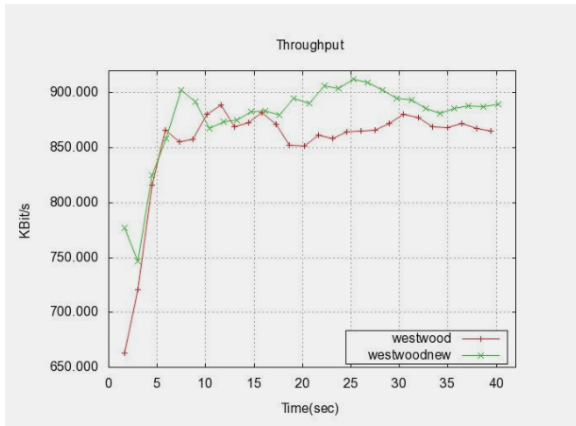
Automation and plotting support

- Separate runner scripts were used for sanity runs and full OAT runs.
- Plotting scripts were created to automatically generate required comparison graphs into structured output directories.
- Additional focused plotting scripts were created for specific two-variant and three-variant RTT/queue analyses.

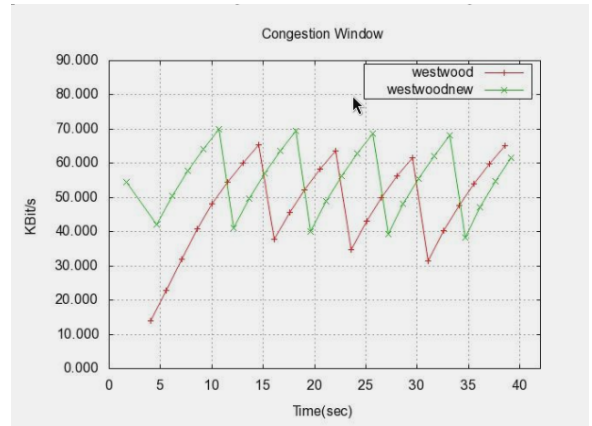
d. Results with graphs

Part 1: TCP time-series comparison

(i) Base paper graphs



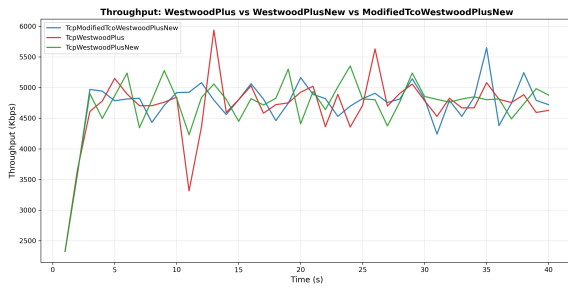
(a) Throughput versus time



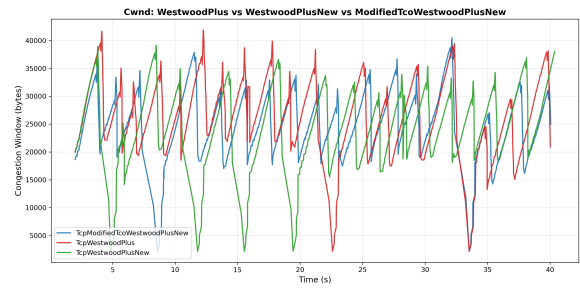
(b) Congestion Window versus time

Figure 1: Base paper graphs.

(ii) My modification graphs



(a) Throughput versus time

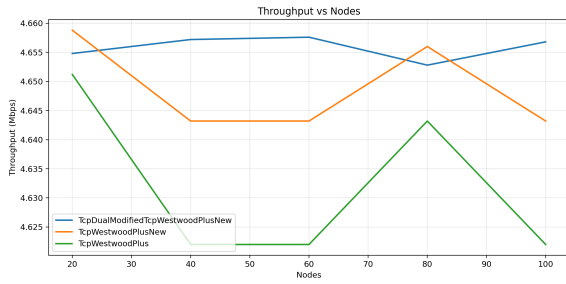


(b) Congestion Window versus time

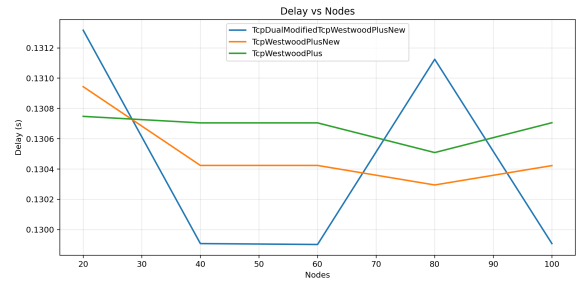
Figure 2: My modification graphs.

Part 2: Wired network

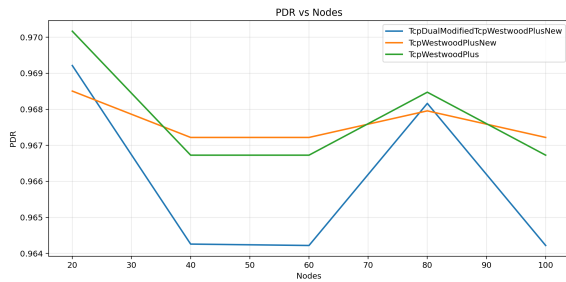
(i) Variation with nodes



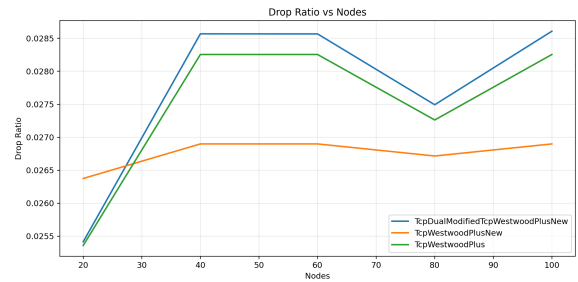
(a) Throughput vs node



(b) Delay vs node



(c) Packet delivery ratio vs node



(d) Packet drop ratio vs node

Figure 1: Wired network results (variation with nodes).

(ii) Variation with flows

(iii) Variation with packet rate

Part 3: Wireless network

(i) Variation with nodes

(ii) Variation with flows

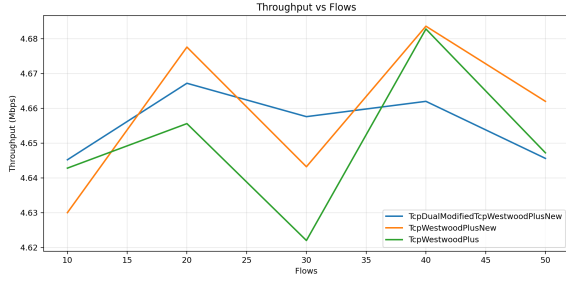
(iii) Variation with packet rate

(iv) Variation with coverage area

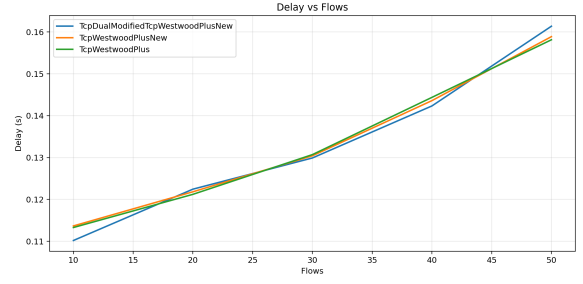
(v) Variation with energy consumption

e. Extra Metrics

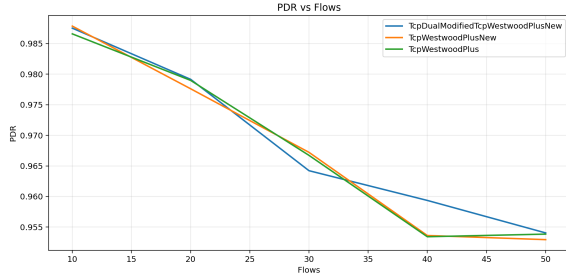
In our case, observing Queue size and RTT over time is very important. So we calculated those 2 metrics extra.



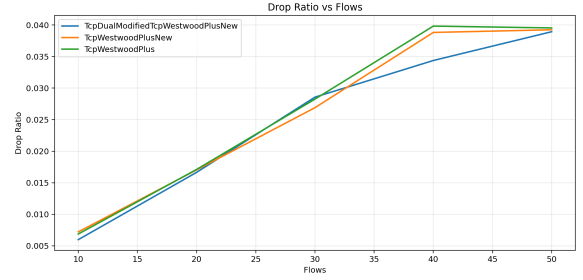
(a) Throughput vs flows



(b) Delay vs flows

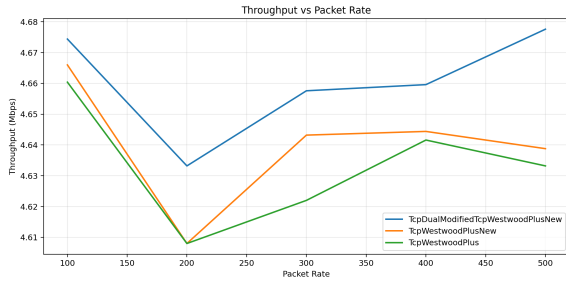


(c) Packet delivery ratio vs flows

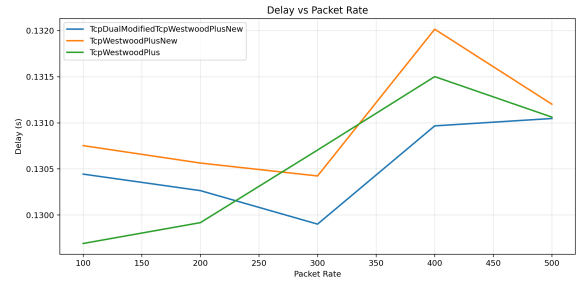


(d) Packet drop ratio vs flows

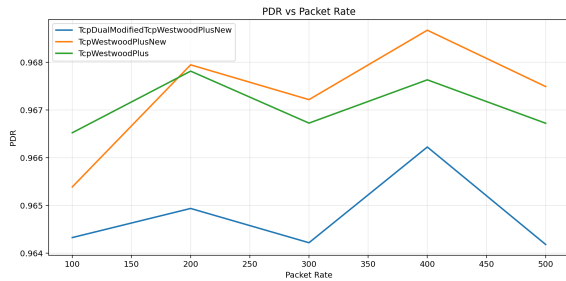
Figure 2: Wired network results (variation with flows).



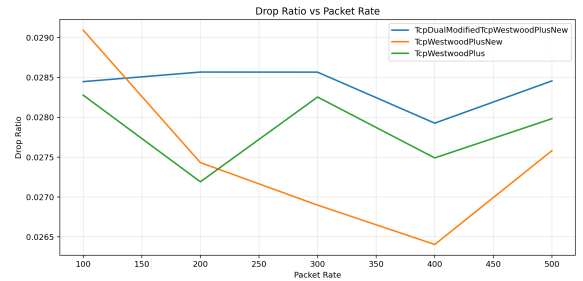
(a) Throughput vs packet rate



(b) Delay vs packet rate

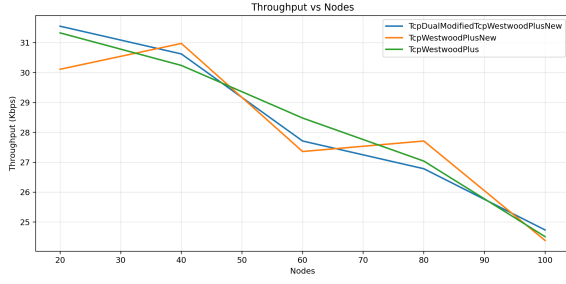


(c) Packet delivery ratio vs packet rate

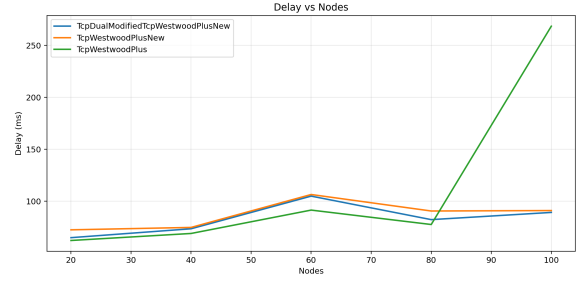


(d) Packet drop ratio vs packet rate

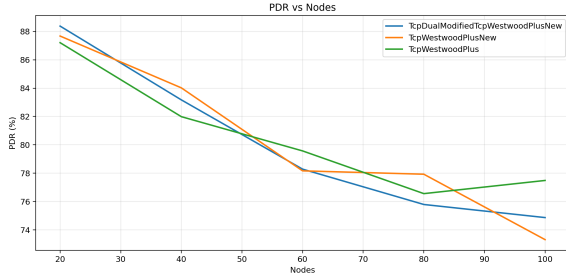
Figure 3: Wired network results (variation with packet rate).



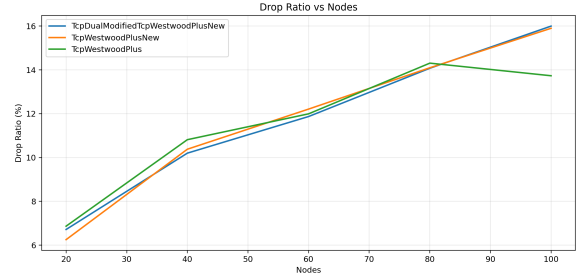
(a) Throughput vs node



(b) Delay vs node

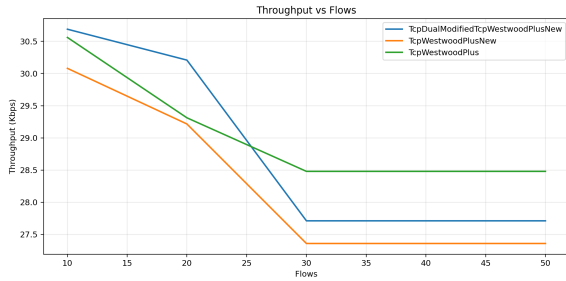


(c) Packet delivery ratio vs node

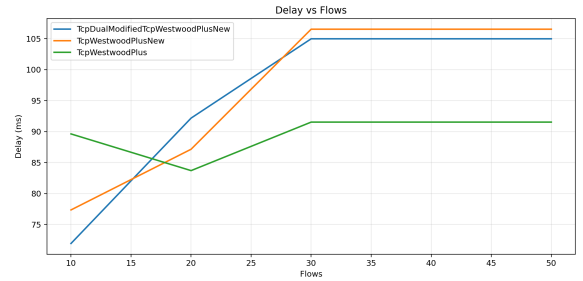


(d) Packet drop ratio vs node

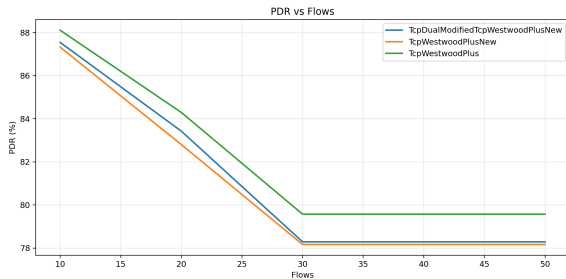
Figure 4: Wireless network results (variation with nodes).



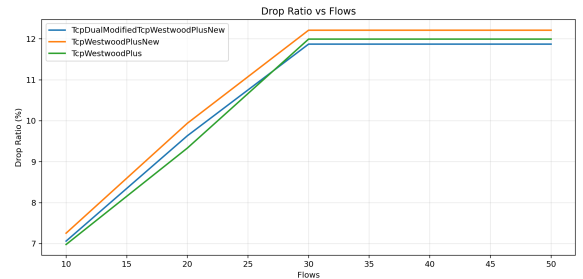
(a) Throughput vs flows



(b) Delay vs flows

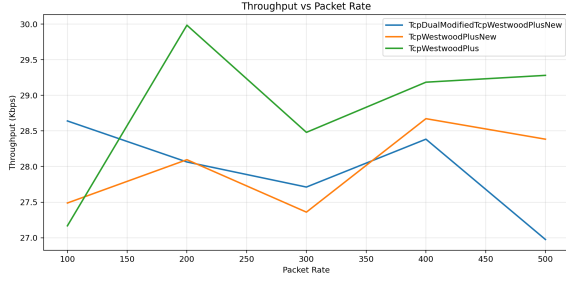


(c) Packet delivery ratio vs flows

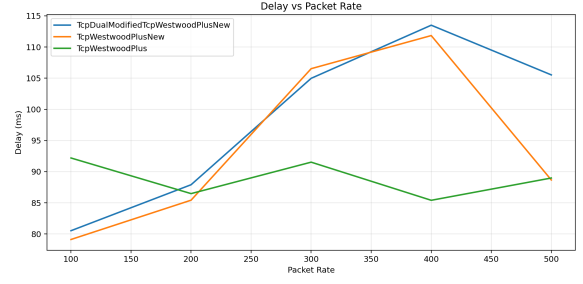


(d) Packet drop ratio vs flows

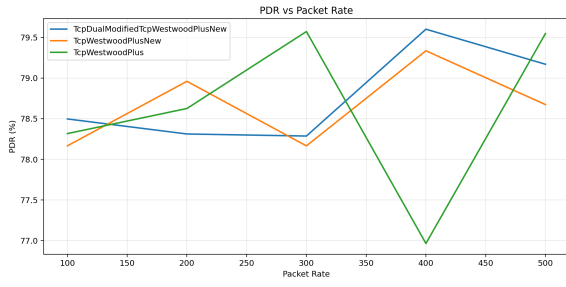
Figure 5: Wireless network results (variation with flows).



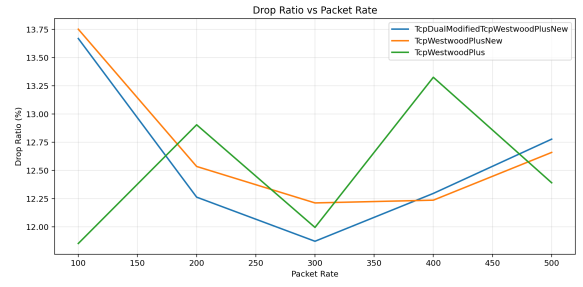
(a) Throughput vs packet rate



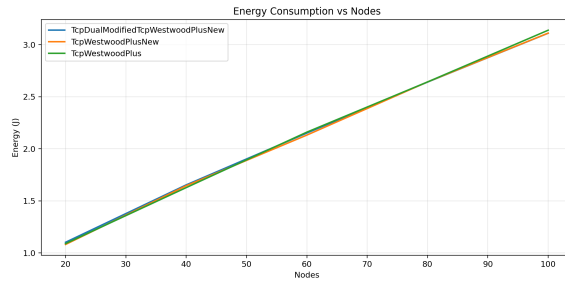
(b) Delay vs packet rate



(c) Packet delivery ratio vs packet rate

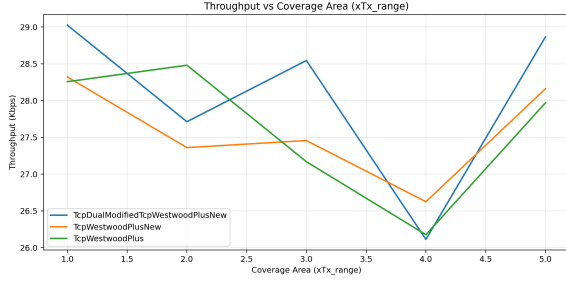


(d) Packet drop ratio vs packet rate

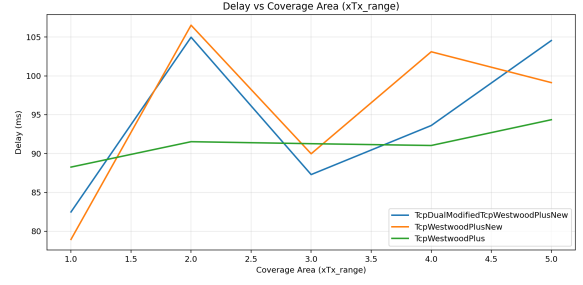


(e) Energy Consumption vs Nodes

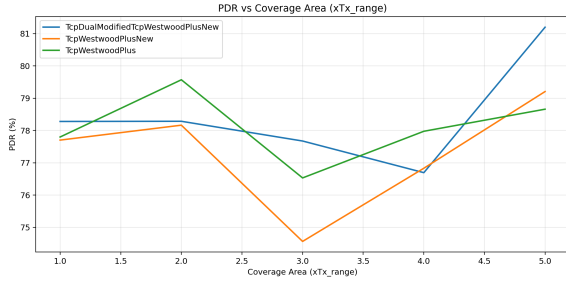
Figure 6: Wireless network results (variation with packet rate).



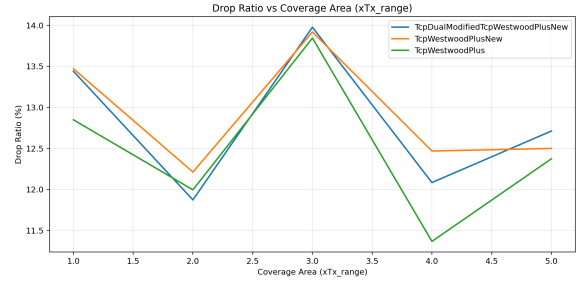
(a) Throughput vs Coverage



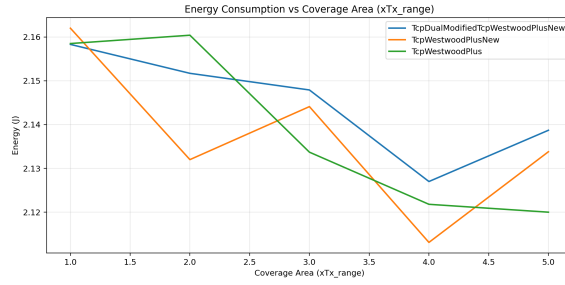
(b) Delay vs Coverage



(c) Packet delivery ratio vs Coverage

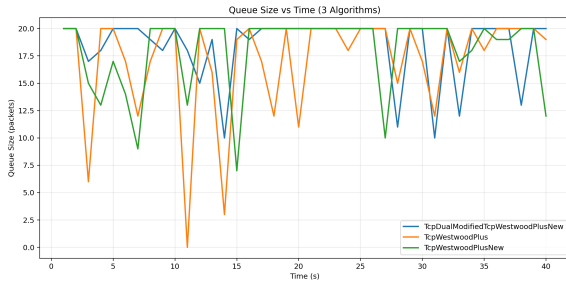


(d) Packet drop ratio vs Coverage

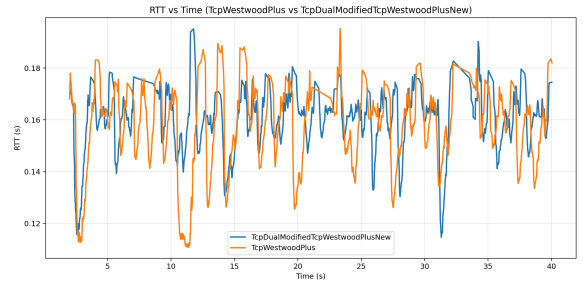


(e) Energy Consumption vs Coverage

Figure 7: Wireless network results (variation with packet rate).



(a) Queue Size vs Time



(b) RTT vs Time

Figure 8: Extra metrics (queue size and RTT over time).

f. Summary Findings

Base: TCPWestwoodPlus

Paper’s Modification: TCPWestwoodPlusNew

My Modification: TCPDualModifiedTCPWestwoodPlusNew

Wired Setup In wired scenarios, the proposed modification consistently outperforms both the base TCPWestwoodPlus and the paper’s TCPWestwoodPlusNew across most performance metrics. Throughput remains within a narrow range of 4.62–4.66 Mbps, indicating stable behavior across all variants. However, the DualModified version achieves slightly higher throughput at moderate to high flow counts, particularly around 300–400 packets per second.

Additionally, the proposed method demonstrates small but consistent improvements in end-to-end delay and packet delivery ratio (PDR), especially when the number of flows exceeds 30. Notably, its PDR degrades more gracefully compared to the other variants, remaining above 96% overall.

Wireless Setup In wireless environments, some metrics remain nearly identical across all variants, while others show noticeable differences. As the number of nodes increases from 20 to 100, throughput decreases for all algorithms. However, the DualModified variant maintains slightly better throughput compared to the base TCPWestwoodPlus.

The most significant improvement is observed in delay performance. At higher node densities (e.g., 100 nodes), DualModified achieves lower end-to-end delay due to proportional congestion window (cwnd) reduction, which helps prevent excessive queue buildup in noisy wireless channels.

However, the packet delivery ratio exhibits a different trend. At high node counts, both DualModified and TCPWestwoodPlusNew perform worse than the base TCPWestwoodPlus.

Varying Flow Conditions Under varying flow counts, DualModified achieves higher throughput than the other variants up to approximately 40 flows. Beyond this point, the base TCPWestwoodPlus provides more stable throughput.

In terms of delay, DualModified clearly outperforms both alternatives, achieving significantly lower latency. When varying packet rates, it reaches peak throughput at around 300 packets per second, although it exhibits higher oscillation at extreme rates.

Energy consumption remains identical across all variants and node configurations, indicating that the proposed modifications introduce no additional power overhead at the wireless layer.

g. Reflection on Findings

Three major observations emerge from the experiments:

1. The proposed modification (TCPDualModifiedTCPWestwoodPlusNew) outperforms both the base TCPWestwoodPlus and the paper’s TCPWestwoodPlusNew in most scenarios.

2. The improvement in throughput and delay often comes at the cost of increased packet loss and reduced packet delivery ratio.
3. In wireless setups, all three algorithms exhibit similar performance overall, except at higher packet rates where the base TCPWestwoodPlus demonstrates greater stability.

These observations align well with the expected theoretical behavior. The proposed modification increases the aggressiveness of congestion window growth, allowing more packets to be injected into the network. However, since the bottleneck link capacity remains fixed, this leads to queue buildup and eventual packet drops.

As a result, packet loss increases and packet delivery ratio decreases. Nevertheless, the increased transmission rate compensates for these losses by utilizing available bandwidth more effectively, leading to higher overall throughput. The results under varying packet rates further support this interpretation, as the aggressive cwnd growth enables better performance at higher traffic rates.

In wireless networks, the behavior remains largely consistent with the above reasoning, except under high packet rate conditions where the base TCPWestwoodPlus achieves more stable throughput, albeit with reduced packet delivery ratio.

Energy consumption follows a linear trend with respect to node count for all three algorithms, as expected. However, under varying coverage scenarios, the proposed modification consumes slightly more power due to increased packet loss and the associated retransmission overhead.